

# Rustem Islamov

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## EDUCATION

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| <b>PhD in Computer Science</b><br><a href="#">University of Basel</a> , Supervisor: <a href="#">Aurelien Lucchi</a>                                                                                                                                                 | Oct. 2023 – Pres.<br>Basel, Switzerland       |
| <b>Master of Science in Applied Mathematics</b><br><a href="#">Institut Polytechnique de Paris</a><br>Supervisor: <a href="#">Olivier Fercoq</a> , Thesis supervisor: <a href="#">Dan Alistarh</a><br>GPA: 17.65/20, <a href="#">transcript of records</a>          | Sept. 2021 – Aug. 2023<br>Palaiseau, France   |
| <b>Bachelor of Science in Applied Mathematics and Physics</b><br><a href="#">Moscow Institute of Physics and Technology</a><br>Supervisor: <a href="#">Peter Richtárik</a><br>GPA: 4.95/5 (9.27/10), Top 3 at the department, <a href="#">transcript of records</a> | Sept. 2017 – July 2021<br>Dolgoprudny, Russia |

## RESEARCH INTERESTS

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Deep Learning (loss landscape of neural networks), Optimization (deep learning optimizers, differential privacy and robustness in distributed optimization, model quantization, computation- and memory-efficient methods)

## INTERNSHIPS AND RESEARCH VISITS

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| <b>Research Visit in the group of Prof. Eduard Gorbunov</b><br><a href="#">MBZUAI</a> , Host: <a href="#">Eduard Gorbunov</a>                                          | Dec. 2025 – Feb. 2026<br>Abu Dhabi, UAE                                |
| <b>Internship at Distributed Algorithms and Systems Lab</b><br><a href="#">IST Austria</a> , Supervisors: <a href="#">Mher Safaryan</a> , <a href="#">Dan Alistarh</a> | Apr. 2023 – Oct. 2023<br>Klosterneuburg, Austria                       |
| <b>Internship at Machine Learning and Optimization Lab</b><br><a href="#">EPFL</a> , Supervisors: <a href="#">Hadrien Hendrikx</a> , <a href="#">Martin Jaggi</a>      | Apr. 2022 – Aug. 2022<br>Lausanne, Switzerland                         |
| <b>Internship at Optimization and Machine Learning Lab</b><br><a href="#">KAUST</a> , Supervisor: <a href="#">Peter Richtárik</a>                                      | Mar. 2021 – Aug. 2021<br>Jul. 2020 – Dec. 2020<br>Thuwal, Saudi Arabia |

## SCHOLARSHIPS, HONORS AND AWARDS

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| <b>Top Reviewer for NeurIPS 2025, ICLR 2026</b><br>Free registration for the conference                                            | 2025-2026                                        |
| <b>University of Basel Travel Fund</b><br>Covers travel costs at EUROPT 2025; merit-based                                          | Aug. 2025                                        |
| <b>ICLR 2025 Financial Assistance</b><br>Cover registration fee; merit-based                                                       | Apr. 2025                                        |
| <b>NeurIPS@Paris Workshop Travel Grant</b><br>Covers the travel expenses during the workshop; merit-based                          | Dec. 2024                                        |
| <b>French Embassy Scholarship</b><br>Given to students enrolled to French universities with high academic achievements             | Sept. 2022 – May. 2023                           |
| <b>PhD Track Excellence Scholarship</b><br>IP Paris awards merit-based excellence scholarships for students enrolled in PhD tracks | Sept. 2021 – Mar. 2022<br>Sept. 2022 – Mar. 2023 |
| <b>Increased State Academic Scholarship</b><br>Given to 4 year Bachelor and Master students at MIPT with scientific achievements   | Feb. 2021 – June 2021<br>Sept. 2020 – Jan. 2021  |
| <b>Abramov scholarship</b><br>Given to 1-3 year students with the best grades at MIPT                                              | Sept. 2017 – June 2020                           |

## PUBLICATIONS (2 NEURIPS, 7 ICML, 4 ICLR, 3 AISTATS, 3 TMLR)

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21. [R. Islamov](#), [R. Machacek](#), [A. Lucchi](#), [A. Silveti-Falls](#), [E. Gorbunov](#), [V. Cevher](#), **On the Role of Batch Size in Stochastic Conditional Gradient Methods**, [In Proc. of 43rd International Conference on Machine Learning](#), 2026.

20. [R. Islamov](#), M. Crawshaw, J. Cohen, R. Gower, **Non-Euclidean Gradient Descent Operates at the Edge of Stability**, In [Proc. of 43rd International Conference on Machine Learning](#), (**Spotlight**), 2026.
19. E. M. Compagnoni, A. Stanghellini, [R. Islamov](#), A. Lucchi, A. Koloskova. **Adaptive Methods Are Preferable in High Privacy Settings: An SDE Perspective**, in [Proc. of the 14th International Conference on Learning Representations](#), 2026.
18. Foivos Alimisis\*, [R. Islamov](#)\*, A. Lucchi (\*equal contribution, alphabetical order). **Why Do We Need Warm-up? A Theoretical Perspective**, In [Proc. of 43rd International Conference on Machine Learning](#), 2025.
17. [R. Islamov](#), N. Ajroldi, A. Orvieto, A. Lucchi. **Enhancing Optimizer Stability: Momentum Adaptation of NGN Step-size**, in [Advances in Neural Information Processing Systems](#), 2025.
16. E. M. Compagnoni, [R. Islamov](#), A. Orvieto, E. Gorbunov. **On the Interaction of Noise, Compression Role, and Adaptivity under  $\epsilon$ -Smoothness: An SDE-based Approach**, In [Proc. of 43rd International Conference on Machine Learning](#), 2025.
15. [R. Islamov](#), Y. As, I. Fatkhullin. **Safe-EF: Error Feedback for Non-smooth Constrained Optimization**, In [Proc. of 42nd International Conference on Machine Learning](#), 2025.
14. [R. Islamov](#), S. Horváth, A. Lucchi, P. Richtárik, E. Gorbunov. **Double Momentum and Error Feedback for Clipping with Fast Rates and Differential Privacy**, [arXiv preprint arXiv: 2502.11682](#), 2025.
13. E. M. Compagnoni, [R. Islamov](#), F. N. Proske, A. Lucchi. **Unbiased and Sign Compression in Distributed Learning: Comparing Noise Resilience via SDEs**, in [Proc. of the 28th International Conference on Artificial Intelligence and Statistics](#), (**Oral**), 2025.
12. E. M. Compagnoni, T. Liu, [R. Islamov](#), F. N. Proske, A. Orvieto, A. Lucchi. **Adaptive Methods through the Lens of SDEs: Theoretical Insights on the Role of Noise**, in [Proc. of the 13th International Conference on Learning Representations](#), 2024.
11. [R. Islamov](#), N. Ajroldi, A. Orvieto, A. Lucchi. **Loss Landscape Characterization of Neural Networks without Over-Parametrization**, in [Advances in Neural Information Processing Systems](#), 2024.
10. [R. Islamov](#)\*, Y. Gao\*, S. Stich(\*equal contribution). **Towards Faster Decentralized Stochastic Optimization with Communication Compression**, in [Proc. of the 13th International Conference on Learning Representations](#), 2024.
9. Y. Gao\*, [R. Islamov](#)\*, S. Stich (\*equal contribution). **EControl: Fast Distributed Optimization with Compression and Error Control**, in [Proc. of the 12th International Conference on Learning Representations](#), 2023.
8. [R. Islamov](#), M. Safaryan, D. Alistarh. **AsGrad: A Sharp Unified Analysis of Asynchronous-SGD Algorithms**, in [Proc. of the 27th International Conference on Artificial Intelligence and Statistics](#), 2023.
7. K. Mishchenko, [R. Islamov](#), E. Gorbunov, S. Horváth. **Partially Personalized Federated Learning: Breaking the Curse of Data Heterogeneity**, [Transactions on Machine Learning Research](#), 2023.
6. S. Khirirat, E. Gorbunov, S. Horváth, [R. Islamov](#), F. Karay, P. Richtárik. **Clip21: Error Feedback for Gradient Clipping**, [arXiv preprint arXiv: 2305.18929](#), 2023.
5. M. Makarenko, E. Gasanov, [R. Islamov](#), A. Sadiev, P. Richtárik. **Adaptive Compression for Communication-Efficient Distributed Training**, [Transactions on Machine Learning Research](#), 2022.
4. [R. Islamov](#), X. Qian, S. Hanzely, M. Safaryan, P. Richtárik. **Distributed Newton-Type Methods with Communication Compression and Bernoulli Aggregation**, [Transactions on Machine Learning Research](#), 2022.
3. X. Qian, [R. Islamov](#), M. Safaryan, P. Richtárik. **Basis Matters: Better Communication-Efficient Second Order Methods for Federated Learning**, in [Proc. of the 25th International Conference on Artificial Intelligence and Statistics](#), 2022.
2. M. Safaryan, [R. Islamov](#), X. Qian, P. Richtárik. **FedNL: Making Newton-Type Methods Applicable to Federated Learning**, In [Proc. of 39th International Conference on Machine Learning](#), 2022.
1. [R. Islamov](#), X. Qian, P. Richtárik. **Distributed Second Order Methods with Fast Rates and Compressed Communication**, In [Proc. of 38th International Conference on Machine Learning](#), 2021.

REVIEWING AND TEACHING

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|-------------------------------------------------------------------------------------------------|----------------------------|
| <b>Teaching Assistant</b> for <a href="#">Continuous Optimization</a> course                    | Spring Semesters 2024-2026 |
| <b>Reviewer</b> for <a href="#">Transactions on Machine Learning Research (TMLR)</a>            | 2023-2026                  |
| <b>Reviewer</b> for <a href="#">Conference on Neural Information Processing Systems</a>         | 2024-2025                  |
| <b>Reviewer</b> for <a href="#">International Conference on Machine Learning (ICML)</a>         | 2024-2026                  |
| <b>Reviewer</b> for <a href="#">Journal on Machine Learning Research (JMLR)</a>                 | 2024-2025                  |
| <b>Reviewer</b> for <a href="#">Journal of Parallel and Distributed Computing (JPDC)</a>        | 2024                       |
| <b>Reviewer</b> for <a href="#">Journal of Optimization Theory and Applications (JOTA)</a>      | 2023                       |
| <b>Reviewer</b> for <a href="#">Artificial Intelligence and Statistics Conference (AISTATS)</a> | 2023-2026                  |
| <b>Reviewer</b> for <a href="#">Neural Information Processing Systems (NeurIPS)</a>             | 2024-2026                  |
| <b>Reviewer</b> for <a href="#">International Conference on Learning Representations (ICLR)</a> | 2024-2026                  |

TALKS AND POSTERS

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| <b>Invited Talk</b> at <a href="#">EUROPT</a> Conference, Links: <a href="#">paper</a>                                                        | 29 June-2 July, 2025 |
| <b>Invited Talk</b> at <a href="#">Rising Stars in AI Symposium at KAUST</a> workshop, Links: <a href="#">paper</a> , <a href="#">slides</a>  | 21 February, 2024    |
| <b>Talk</b> at <a href="#">NTDS</a> workshop, Links: <a href="#">paper</a>                                                                    | 24 October, 2023     |
| <b>Invited Talk</b> at <a href="#">CISPA</a> , Links: <a href="#">paper</a>                                                                   | 16 March, 2023       |
| <b>Talk</b> at <a href="#">ETH AI Center</a> Symposium for PhD fellows, Links: <a href="#">paper</a>                                          | 9-10 February, 2023  |
| <b>Prerecorded Talk</b> at <a href="#">KAUST Conference on Artificial Intelligence</a> , Links: <a href="#">video</a> , <a href="#">paper</a> | 28 April, 2021       |

TECHNICAL SKILLS

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| <b>Programming Languages:</b> Python (NumPy, Matplotlib, PyTorch, Pandas), C++, LaTeX            |
| <b>Mathematics:</b> Calculus, Linear Algebra, Probability Theory, Convex Analysis, Deep Learning |

LANGUAGES

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| <b>Russian:</b> Native                        |
| <b>English:</b> <a href="#">Advanced (C1)</a> |
| <b>French:</b> Elementary (A2)                |

HOBBIES AND INTERESTS

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| Football, former member of student football team |
| Travelling, hiking, photo shooting               |